

Frontend: LLM Generation Pipeline

Natural
Language

user input

LLM

llama3.1

LIR

DSL

Compiler

deterministic

Verifier

static check

Bytecode

varint

repaired LIR

Backend: MCU Execution & Runtime Protection

Transmission

serial/WiFi

LVM

ESP8266

Hardware I/O

relay/sensor

Feedback

fault report

Repair Loop

fix & retry

• Cap. Gating

• Step Limit

• Fault Audit

• Closed-loop Verify